

Principles of Geographic Information Systems

MODULE-2: *Data management and processing systems*



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Certificate

This is to certify that the e-book titled “PRINCIPLES OF GEOGRAPHIC INFORMATION SYSTEMS” comprises all elementary learning tools for a better understating of the relevant concepts. This e-book is comprehensively compiled as per the predefined eight parameters and guidelines.



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Unit II

Contents:

Spatial Data Analysis

- Hardware and software trends
- Geographic information systems
- Stages of spatial data handling
- Database management systems
- GIS and spatial databases

Recommended Books

- Principles of Geographic Information Systems- An Introductory Text Book Editors: Otto Huisman and Rolf A. Fourth 2009
- Principles of Geographic Information Systems P.A Burrough and R.A.McDonnell Oxford University Press Third 1999
- Fundamentals of Spatial Information Systems, R.Laurini and D. Thompson, Academic Press 1994
- Fundamentals of Geographic Information Systems Michael N.Demers Wiley Publications Fourth 2009
- Introduction to Geographic Information Systems Chang Kang-tsung (Karl), McGrawHill Any above 3rd Edition 2013 7th Edition
- GIS Fundamentals: A First Text on Geographic Information Systems Paul Bolsatd XanEdu Publishing Inc 5th Edition

Prerequisites and Linking:

Unit II	Pre-requisites	Sem. II	Sem. III	Sem. IV	Sem. V	Sem. VI
Data management and processing systems		DBMS	CGA			Projects

Chapter 3

Data management and processing systems

The ability to manage and process spatial data is a critical component for any functioning GIS. Data processing systems refer to hardware and software components which can process, store and transfer data.

3.1 Hardware and software trends

Advances in computer hardware seem to take place at an ever-increasing rate. Every several months, a faster, more powerful processor generation replaces the previous one today's handheld computers is a multiple of the performance that the first PC

Handheld PC's had when it was introduced in the early 1980's. In fact, current PCs have orders

of magnitude more memory and storage capacity than the so-called minicomputers of 25 years ago. Computers are also becoming increasingly affordable. Hand-held computers are now commonplace in business and personal use, equipping field surveyors with powerful tools, complete with GPS capabilities for instantaneous georeferencing. To support these hardware trends, software providers continue to produce application programs and operating systems that, while providing a lot more functionality, also consume significantly more memory. In general, software technology has developed somewhat slower and often cannot fully utilize the possibilities offered by the exponentially growing hardware capabilities. Existing software obviously performs better when run on faster computers.

Today almost any computer on Earth can connect to some network, and contact computers virtually anywhere else, allowing fast and reliable exchange of (spatial) data. Mobile phones are more and more frequently The UMTS protocol (Universal Mobile Telecommunications System), allows digital communication of text, audio, and video at a rate of approximately 2 Mbps. The new HSDPA protocol offers up to 10 times this speed. Looking at these developments it is clear that the combination of a GPS receiver, a portable computer and mobile phone has already dramatically changed our world, certainly so for out-of-office activities of Earth science professionals.

Wireless LANs (Local Area Networks), under the so-called WiFi standard, nowadays offer a bandwidth of up to 108 Mbps on a single connection point, to be shared between computers. They are more and more used for constructing a computer network in office buildings and in private homes.

Standard 'Dial-up' telephone modems allow rates up to 56 kbps. Digital telephone links (ISDN) support much higher rates: up to 1.5 Mbps. ADSL technology widely available through telephone companies on standard copper-wire networks supports transfer rates anywhere between 2 and 20 Mbps towards the customer (down-stream), and between 1 and 8 Mbps towards the network (upstream) depending on the internet provider and quality of the network infrastructure.

3.2 Geographic information systems

GIS provides a range of capabilities to handle georeferenced data, including:

1. Data capture and preparation
2. Data management (storage and maintenance)
3. Data manipulation and analysis, and
4. Data presentation.

Planning projects require data sources, both spatial and non-spatial, from different national institutes, like national mapping agencies, geological, soil, and forest survey institutes, and national census bureaus. The data sources obtained may be from different time periods, and the spatial data may be in different scales or projections. With the help of a GIS, the spatial data can be stored in digital form in world coordinates. This makes scale transformations unnecessary, and the conversion between map projections can be done easily with the software.

3.2.1 GIS software

GIS can be a data store (i.e. a system that stores spatial data), a toolbox, a technology, an information source or a field of science. The main characteristics of a GIS software package are its analytical functions that provide means for deriving new geoinformation from existing spatial and attribute data.

The use of tools for problem solving is one thing, but the production of these tools is something quite different. Not all tools are equally well-suited for a application, and they can be improved and perfected to better serve a particular need or application. The discipline of geographic information science is driven using our GIS tools.

Some GISs have traditionally focused more on support for raster-based functionality, others more on (vector-based) spatial objects. We can safely state that any package that provides support for only raster's or only objects, is not a complete GIS. Well-

known, full-fledged GIS packages include ILWIS, Intergraph's GeoMedia, ESRI's ArcGIS, and MapInfo from Map-Info Corp.

There is no GIS package which is necessarily 'better' than another one: this depends on factors such as the intended application, and the expertise of its user. ILWIS's traditional strengths are in raster processing and scientific spatial data analysis, especially in project-based GIS applications. Intergraph, ESRI and MapInfo products have been known better for their support of vector-based spatial data and their operations, user interface and map production (a bit more typical of institutional GIS applications).

Videio on GIS <https://www.youtube.com/watch?v=-ZFmAAHBfOU>

Video on GIS Software:

[Types of GIS Software's | Types of GIS Software | gis mapping software | Geology GIS - YouTube](#)

3.2.2 GIS architecture and functionality

A GIS consists of several functional components—components which support key GIS functions. These are data capture and preparation, data storage, data analysis, and presentation of spatial data. Figure shows a diagram of these components, with arrows indicating the data flow in the system. For a particular GIS, each of these components may provide many or only a few functions.

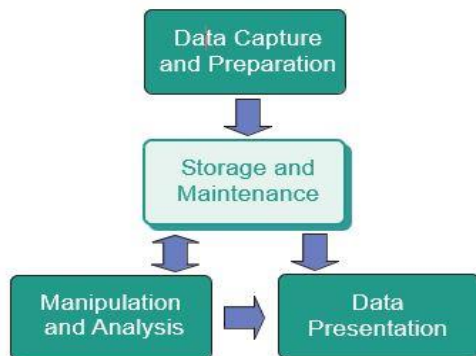


Fig Functional components of a GIS

Video on GIS System Architecture and Components

<https://www.youtube.com/watch?v=4ze4rHXlxrM>

3.2.3 Spatial Data Infrastructure (SDI)

Many organizations are forced to work in a cooperative setting in which geographic information is obtained from, and provided to, partner organizations and the general public. The sharing of spatial data between the various GISs in those organizations is of key importance and aspects of data dissemination, security, copyright and pricing require special attention. The design and maintenance of a Spatial Data Infrastructure (SDI) deals with these issues.

An SDI is defined as “the relevant base collection of technologies, policies and institutional arrangements that facilitate the availability of and access to spatial data”. Fundamental to those arrangements are in a wider sense the agreements between organizations and in the narrow sense, the agreements between software systems on how to share the geographic information.

Standards exist for all facets of GIS, ranging from data capture to data presentation. They are developed by different organizations, of which the most prominent are the International Organization for Standardization (ISO) and the Open Geospatial Consortium (OGC).

Additional learning: To know details about OGC standards.

[Standards \(ogc.org.tw\)](http://ogc.org.tw)

Video on SDI: <https://www.youtube.com/watch?v=Dwd7jZD0sMM>

3.3 Stages of spatial data handling

The functions for capturing data are closely related to the disciplines of surveying engineering, photogrammetry, remote sensing, and the processes of digitizing, i.e. the conversion of analogue data into digital representations.

3.3.1 Spatial data capture and preparation

Traditional techniques for obtaining spatial data, typically from paper sources, included manual digitizing and scanning. Table lists the main methods and devices used for data capture. In recent years there has been a significant increase in the availability and sharing of digital (geospatial) data.

Method	Devices
Manual digitizing	coordinate entry via keyboard digitizing tablet with cursor mouse cursor on the computer monitor (heads-up digitizing)
Automatic digitizing	Scanner
Semi-automatic digitizing	line-following software
Input of available digital data	CD-ROM or DVD-ROM via computer network or internet (including geo-webservices)

Additional learning: Explore the GIS data capturing methods.

[Want to know more about data capture methods in GIS? \(uizentrum.de\)](http://uizentrum.de)

The data, once obtained in some digital format, may not be quite ready for use in the system. This may be because the format obtained from the capturing process is not quite the format required for storage and further use, which means that some type of data conversion is required.

3.3.2 Spatial data storage and maintenance

The way that data is stored plays a central role in the processing and the eventual understanding of that data. In most of the available systems, spatial data is organized in layers by theme and/or scale.

In a GIS, features are represented with their (geometric and non-geometric) attributes and relationships. The geometry of features is represented with primitives of the respective dimension: a windmill probably as a point, an agricultural field as a polygon.

Cells, pixels and voxels:

Vector data types describe an object through its boundary, thus dividing the space into parts that are occupied by the respective objects.

The raster approach subdivides space into (regular) cells, mostly as a square tessellation of dimension two or three. These cells are called either cells or pixels in 2D, and voxels in 3D.

The data indicates for every cell which real world feature it covers, in case it represents a discrete field. In case of a continuous field, the cell holds a representative value for that field.

Raster representation	Vector representation
Advantages	
simple data structure simple implementation of overlays efficient for image processing	efficient representation of topology adapts well to scale changes allows representing networks allows easy association
Disadvantages	
less compact data structure difficulties in representing topology cell boundaries independent	complex data structure overlay more difficult to implement inefficient for image processing more update-intensive

Raster encoding

This simple encoding scheme is known as row ordering. The header of the raster file will typically inform how many rows and columns the raster has, which encoding scheme is used, and what sort of values are stored for each cell. Raster files can be quite big data sets. For computational reasons, it is wise to organize the long list of cell values in such a way that spatially nearby cells are also near to each other in the list.

DBMS and spatial databases

GIS software packages provide support for both spatial and attribute data, i.e. they accommodate spatial data storage using a vector approach, and attribute data using

tables. Historically, however, database management systems (DBMSs) have been based on the notion of tables for data storage. For some time, substantial GIS applications have been able to link to an external database to store attribute data and make use of its superior data management functions. Currently, All major GIS packages provide facilities to link with a DBMS and ex-change attribute data with it. Spatial (vector) and attribute data are still sometimes stored in separate structures, although they can now be stored directly in a spatial database

Data maintenance

Maintenance of (spatial) data can best be defined as the combined activities to keep the data set up-to-date and as supportive as possible to the user community. It deals with obtaining new data, and entering them into the system, possibly replacing outdated data. The purpose is to have an up-to-date stored data set available. After a major earthquake, for instance, we may have to update our road network data to reflect that roads have been washed away ,or have otherwise become impassable.

Updating spatial data stems from the requirements that the data users impose, as well as the fact that many aspects of the real world change continuously. These data updates can take different forms. It may be that a complete, new survey has been carried out, from which an entirely new data set is derived that will replace the current set.

3.3.3 Spatial query and analysis

The most distinguishing parts of a GIS are its functions for spatial analysis, i.e. operators that use spatial data to derive new geoinformation. Spatial queries and process models play an important role in this functionality. One of the keys uses of

GISs has been to support spatial decisions. Spatial decision support systems (SDSS) are a category of information systems composed of a database, GIS software, models, and a so-called knowledge engine which allow users to deal specifically with locational problems.

In a GIS, data are usually grouped into layers (or themes). Usually, several themes are part of a project. The analysis functions of a GIS use the spatial and non-spatial attributes of the data in a spatial database to provide answers to user questions. GIS functions are used for maintenance of the data, and for analyzing the data in order to infer information from it. Analysis of spatial data can be defined as computing new information that provides new insight from the existing, stored spatial data.

Analysis of spatial data can be defined as computing new information that provides new insight from the existing, stored spatial data.

Ex: In mountainous areas this is a complex engineering task with many cost factors, which include the amount of tunnels and bridges to be constructed, the total length of the tarmac, and the volume of rock and soil to be moved. GIS can help to compute such costs on the basis of an up-to-date digital elevation model and soil map

3.3.4 Spatial data presentation

The presentation of spatial data, whether in print or on-screen, in maps or in tabular displays, or as 'raw data', is closely related to the disciplines of cartography, printing and publishing. The presentation may either be an end-product, for example as a printed atlas, or an intermediate product, as in spatial data made available through the internet

Method	Devices
Hard copy	Printer plotter (pen plotter, ink-jet printer, thermal transfer printer, electrostatic plotter)
Soft copy	computer screen
Output of digital data sets	magnetic tape CD-ROM or DVD the Internet

3.4 Database management systems

Designing a database is not an easy task. Firstly, one must consider carefully what the database purpose is, and who its users will be. Secondly, one needs to identify the available data sources and define the format in which the data will be organized within the database. This format is usually called the database structure. Lastly, data can be entered into the database. It is important to keep the data up-to-date, and it is therefore wise to set up the processes for this, and make someone responsible for regular maintenance of the database.

3.4.1 Reasons for using a DBMS

There are various reasons why one would want to use a DBMS for data storage and processing.

A DBMS supports the storage and manipulation of very large data sets.

A DBMS can be instructed to guard over data correctness.

A DBMS supports the concurrent use of the same data set by many users.

A DBMS provides a high-level, declarative query language.

A DBMS always includes data backup and recovery functions to ensure data availability.

A DBMS allows the control of data redundancy.

3.4.2 Alternatives for data management

The decision whether to use a DBMS will depend, among other things, on how much data there is or will be, what type of use will be made of it, and how many users might be involved.

when the data set is small, its use relatively simple, and with just one user—we might use simple text files, and a text processor. Think of a personal address book as an example, or a small set of simple field observations. Text files offer no support for data analysis whatsoever, except perhaps in alphabetical sorting.

If our data set is still small and numeric by nature, and we have a single type of use in mind, a spreadsheet program will suffice. This might be the case if we have a number of field observations with measurements that we want to prepare for statistical analysis, for example. However, if we carry out region- or nation-wide censuses, with many observation stations and/or field observers and all sorts of different measurements, one quickly needs a database to keep track of all the data.

Video on Introduction to database <https://www.youtube.com/watch?v=d11viALaCvA>

3.4.3 The relational data model

A data model is a language that allows the definition of:

The structures that will be used to store the base data,

The integrity constraints that the stored data must obey at all moments in time, and

The computer programs used to manipulate the data.

For the relational data model, the structures used to define the database are attributes, tuples and relations. Computer programs either perform data extraction from the database without altering it, in which case we call them queries, or they change the database contents, and we speak of updates or transactions

<u>PrivatePerson</u>	<u>TaxId</u>	<u>Surname</u>	<u>BirthDate</u>
	101-367	Garcia	10/05/1952
	134-788	Chen	26/01/1964
	101-490	Fakolo	14/09/1931

<u>Parcel</u>	<u>Pld</u>	<u>Location</u>	<u>Area</u>	<u>Size</u>
	3421	2001		435
	8871	1462		550
	2109	2323		1040
	1515	2003		245

<u>TitleDeed</u>	<u>Plot</u>	<u>Owner</u>	<u>DeedDate</u>
	2109	101-367	18/12/1996
	8871	101-490	10/01/1984
	1515	134-788	01/09/1991
	3421	101-367	25/09/1996

Relations, tuples and attributes

A table or relation is itself a collection of tuples (or records). In fact, each table is a collection of tuples that are similarly shaped.

An attribute is a named field of a tuple, with which each tuple associates a value, the tuple's attribute value.

An attribute's domain is a (possibly infinite) set of atomic values such as the set Of integer number values, the set of real number values

When a relation is created, we need to indicate what type of tuples it will store. This means that we must.

1. Provide a name for the relation,
2. Indicate which attributes it will have, and
3. Set the domain of each attribute.

Finding tuples and building links between them

A key of a relation comprises one or more attributes. A value for these attributes uniquely identifies a tuple.

If we have a value for each of the key attributes, we are guaranteed to find no more than one tuple in the table with that combination of values. It remains possible that there is no tuple for the given combination

Videio on DATABASE RELATIONAL MODEL

<https://www.youtube.com/watch?v=zVwI9jRWRw>

3.4.4 Querying a relational database

The three query operators have some traits in common. First, all of them require input and produce output, and both input and output are relations! This guarantees that the output of one query (a relation) can be the input of another query, and this gives us the possibility to build more and more complex queries

The first operator is Tuple selection works like a filter: it allows tuples that meet the selection condition to pass, and disallows tuples that do not meet the condition

```
SELECT FROM      Parcel
WHERE            AreaSize > 1000
```

A second operator is attribute projection. Attribute projection works like a tuple formatter: it passes through all tuples of the input but reshapes each of them in the same way.

```
SELECT PId, Location FROM Parcel
```

Our third query operator differs from the two above in that it requires two input relations.

The join operator takes two input relations and produces one output relation, gluing two tuples together (one from each input relation), to form a bigger tuple, if they meet a specified condition.

```
SELECT  
  
FROM TitleDeed, Parcel  
  
WHERE TitleDeed.Plot = Parcel.PId
```

UNARY OPERATOR

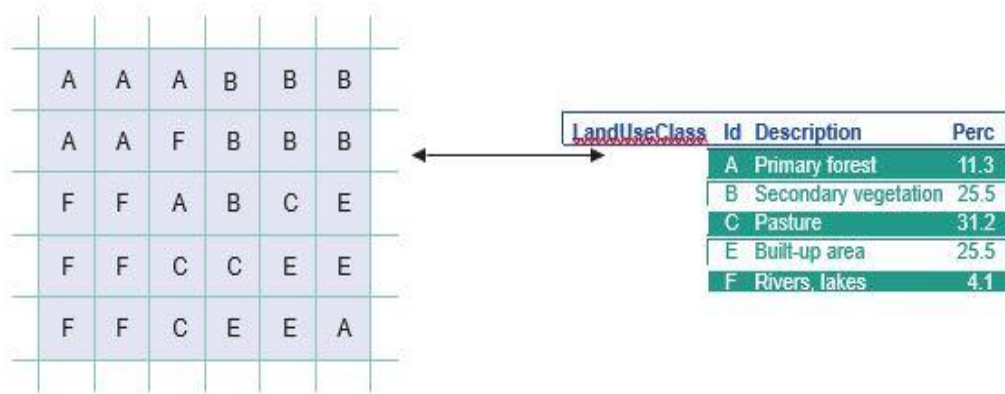
3.5 GIS and spatial databases

3.5.1 Linking GIS and DBMS

GIS software provides support for spatial data and thematic or attribute data. GISs have traditionally stored spatial data and attribute data separately. This required the GIS to provide a link between the spatial data (represented with rasters or vectors), and their non-spatial attribute data. The strength of GIS technology lies in its built-in 'understanding' of geographic space and all functions that derive from this, for purposes such as storage, analysis, and map production.

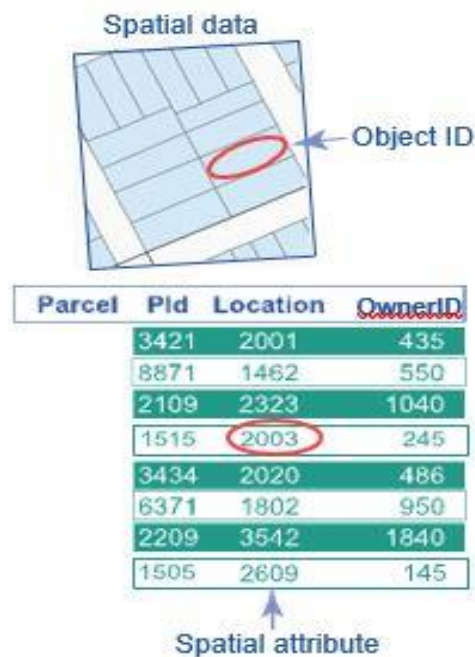
DBMSs offer much better table functionality, since they are specifically designed for this purpose. A lot of the data in GIS applications is attribute data, so it made sense to use a DBMS for it. For this reason, many GIS applications have made use of external DBMSs for data support. In this role, the DBMS serves as a centralized data repository for all users, while each user runs her/his own GIS software that obtains its data from the DBMS. This meant that a GIS had to link the spatial data represented with raster's or vectors, and the attribute data stored in an external DBMS.

With raster representations, each raster cell stores a characteristic value. This value can be used to look up attribute data in an accompanying database table.



With vector representations, our spatial objects—whether they are points, lines or polygons—are automatically given a unique identifier by the system. This identifier is usually just called the object ID or feature ID and is used to link the spatial object (as represented in vectors) with its attribute data in an attribute table.

The ID in the vector system functions as a key, and any reference to an ID value in the attribute database is a foreign key reference to the vector system. For example, in Figure 3.8, parcel is a table with attributes, linked to the spatial objects stored in a GIS by the Location column. Obviously, several tables may make references to the vector system, but it is not uncommon to have some main table for which the ID is actually also the key.



3.5.2 Spatial database functionality

The main problem was that there is additional functionality needed by DBMS in order to process and manage spatial data. As the capabilities of our hardware to process information has increased, so too has the desire for better ways to represent and manage spatial data. During the 1990's, object-oriented and object-relational data models were developed for just this purpose. These extend standard relational models with support for objects, including 'spatial' objects.

Currently, GIS software packages are able to store spatial data using a range of commercial and open source DBMSs such as Oracle, Informix, IBM DB2, Sybase, and PostgreSQL, with the help of spatial extensions. Some GIS software have integrated database 'engines', and therefore do not need these extensions.

ESRI's ArcGIS, for example, has the main components of the MS Access data spatial DBMS base software built-in. This means that the designer of a GIS application can choose whether to store the application data in the GIS or in the DBMS

GIS and spatial databases

Spatial data can be stored in a special database column, known as the geometry column, (or feature or shape, depending on the specific software package), as shown in Figure This means GISs can rely fully on DBMS support for spatial data, making use of a DBMS for data query and storage (and multi-user support), and GIS for spatial functionality. Small-scale GIS applications may not require a multi-user capability and can be supported by spatial data support from a personal database.

Parcel	PId	Geometry	OwnerID
3421		"MULTIPOLYGON(((257462.704979333 464780.750851061,257463.89798...)))"	435
8871		"MULTIPOLYGON(((257409.813950544 464789.91585049,257407.896903...)))"	550
2109		"MULTIPOLYGON(((257785.714911912 464796.839972167,257782.59794...)))"	1040
1515		"MULTIPOLYGON(((257790.672100448 464807.13792585,257788.608078...)))"	245
3434		"MULTIPOLYGON(((257435.527950478 464803.92887633,257428.254887...)))"	486
6371		"MULTIPOLYGON(((257432.476077854 464813.848852072,257433.147910...)))"	950
2209		"MULTIPOLYGON(((257444.888027332 464826.555046319,257446.43201...)))"	1840
1505		"MULTIPOLYGON(((256293.760107491 464935.203846095,256292.00881...)))"	145

Fig. Geometry data stored directly in a spatial database table

Querying a spatial database

A Spatial DBMS provides support for geographic co-ordinate systems and transformations. It also provides storage of the relationships between features, including the creation and storage of topological relationships

```
SELECT R.Name  
  
FROM Restaurants AS R, Hotels as H  
  
WHERE R.Type = "Thai" AND H.name = "Hilton" AND  
  
ST Intersects(R.Geometry, ST Buffer(H.Geometry, 2000))
```

In this case the WHERE clause uses the ST Intersects function to perform a spatial join between a 2000 m buffer of the selected hotel and the selected subset of restaurants. The Geometry column carries the spatial data.

Additional learning: To learn the mapping traits/diversity/species richness at different resolution with code. [Spatial Data Analysis Case Studies \(rspatial.org\)](https://rspatial.org)

Graded Questions

1. What are the stages of GIS
2. Write a note on Spatial Data presentation.
3. State the difference between raster and vector representation.
4. What is DBMS? Explain its reasons.
5. Explain the relational data model using suitable example.
6. How query can be implemented in relational database? What are its three ways?
7. Explain Timestamp based protocol
8. What are the different ways of spatial data capture and preparation? Explain.
9. What is data capture and maintenance? What are the inputs methods and devices used?
10. What are the functional components of GIS

Multiple Choice Questions

1) Full form of SDI is_____

- A. spatial data industry
- B. static data infrastructure

C. Spatial data infrastructure

D. none of them

2) Standard exist for all facets of GIS are _____and _____

A. ISI ,ISO

B.ISO,OGC

C. ISI,OGC

D. none of them

3) Raster approach subdivides space into regular cells, In 2D these cells are called

A. pixel

B. Voxel

C. cell

D. cluster

4) Raster approach subdivides space into regular cells, In 3D these cells are called

A. pixel

B. Voxel

C. cell

D. cluster

5) Raster encoding is also called_____

A. column encoding

B. table encoding

C. row encoding

D. All of them

6) SDSS stands for_____

A. spatial data support system

B. structural data support system

C. Spatial decision support system

D. None of them

7) Primary key is represented in table by using_____

A. underscore(_)

B. hyphen(-)

C. Underline(_____)

D. PK

8) Set of tuples in a relation at some point in time called

A. instance

B. row

C. attribute

D. All of them

9) Spatial Database allow user to _____ the data

A. Store

B. Query

C. Manipulate

D. All of the above

10) Two unary query operator are _____ and _____ -

A. Tuple selection and project selection

B. tuple selection

C. project selection

D. All of the above

11) OGC stands for _____

A. Organization of Geospatial Consortium

B. Origin of GIS Committee

C. Origami of GIS Council

D. All of the above

12) GIS software packages are able to store spatial data using a range of commercial and open source DBMSs.

- A. Oracle
- B. Informix
- C. IBM DB2

D. All of the above

13) _____ representation is complex data structure

A. Vector

- B. Raster
- C. Boolean
- D. Relational

14) The first operator is _____ selection works like a filter: it allows records that meet the selection condition to pass, and disallows tuples that do not meet the condition.

- A. Table
- B. Cell

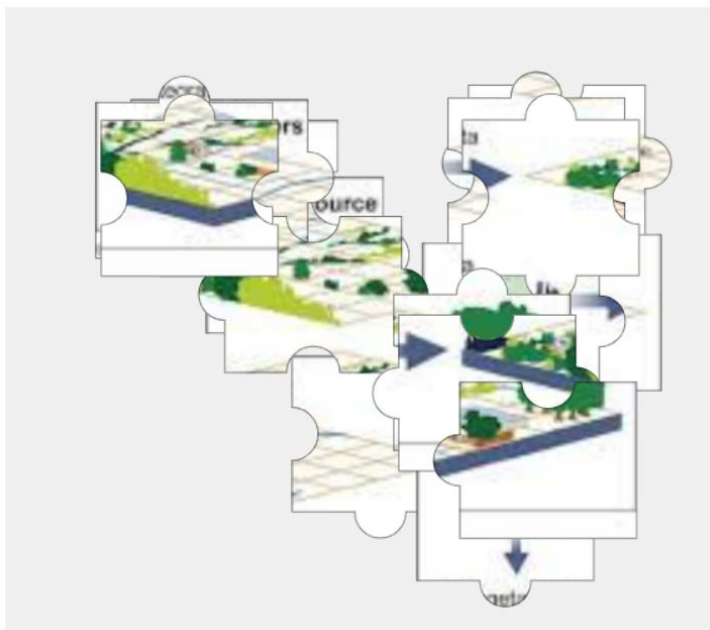
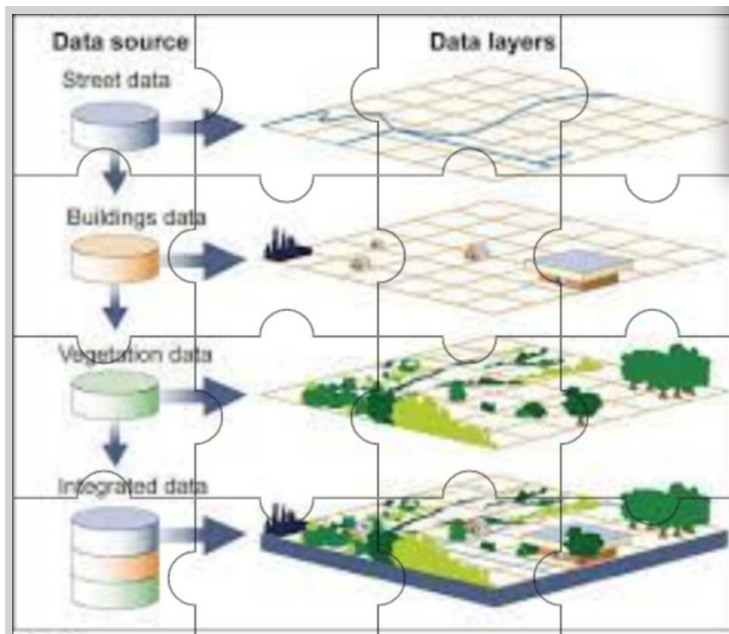
C. Tuple

- D. Column

15) DBMS works at the _____ of GIS software to store and represent data.

- A. Front End
- B. Back End**
- C. Side End
- D. Last End

Additional Activity: Learning by Doing LbD



Click on the link and solve the puzzle

<https://puzzel.org/en/jigsaw/play?p=-MtytgfSBcvkaHQPv4pw>
